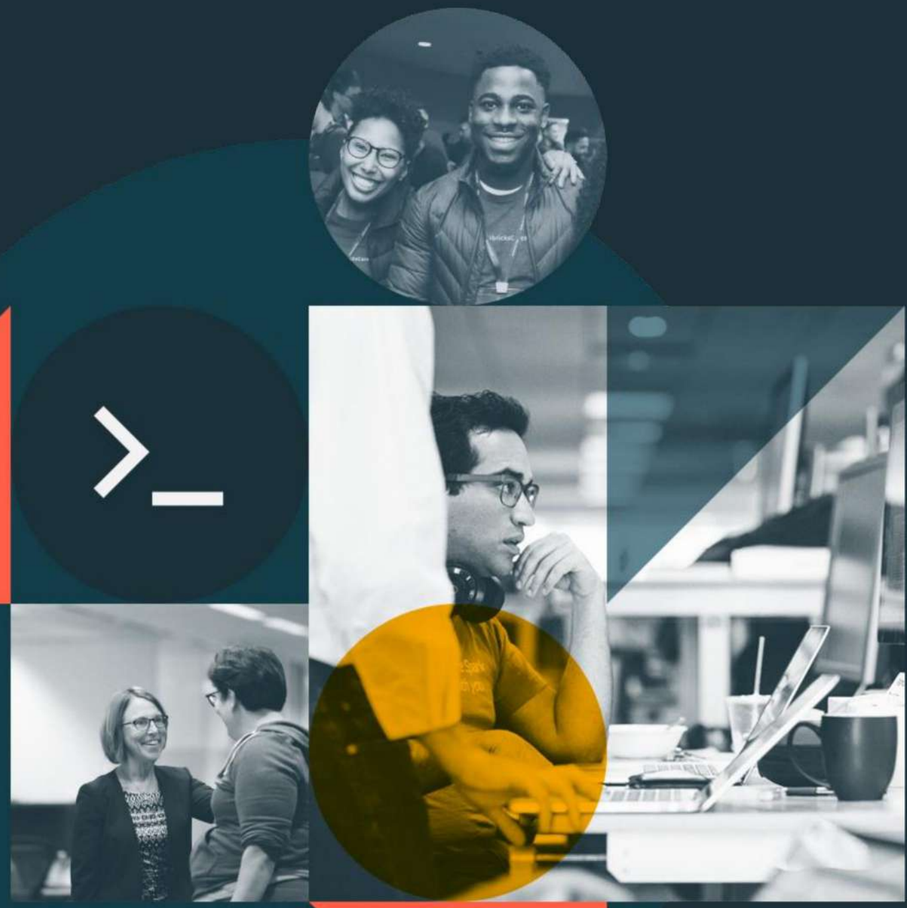
Transform Data with Spark



Module 02

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Module Objectives

Transform Data with Spark

1. Extract data from a variety of file formats and data sources using Spark
2. Apply a number of common transformations to clean data using Spark
3. Reshape and manipulate complex data using advanced built-in functions in Spark
4. Leverage UDFs for reusable code and apply best practices for performance in Spark

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Module Agenda

Transform Data with Spark

Data Objects in the Lakehouse

DE 2.1 - Querying Files Directly

DE 2.2 - Options for External Sources

DE 2.3L - Extract Data Lab

DE 2.4 - Cleaning Data

DE 2.5 - Complex Transformations

DE 2.6L - Reshape Data Lab

DE 2.7A – SQL UDFs and Control Flow

DE 2.7B - Python UDFs

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Data Objects in the Lakehouse

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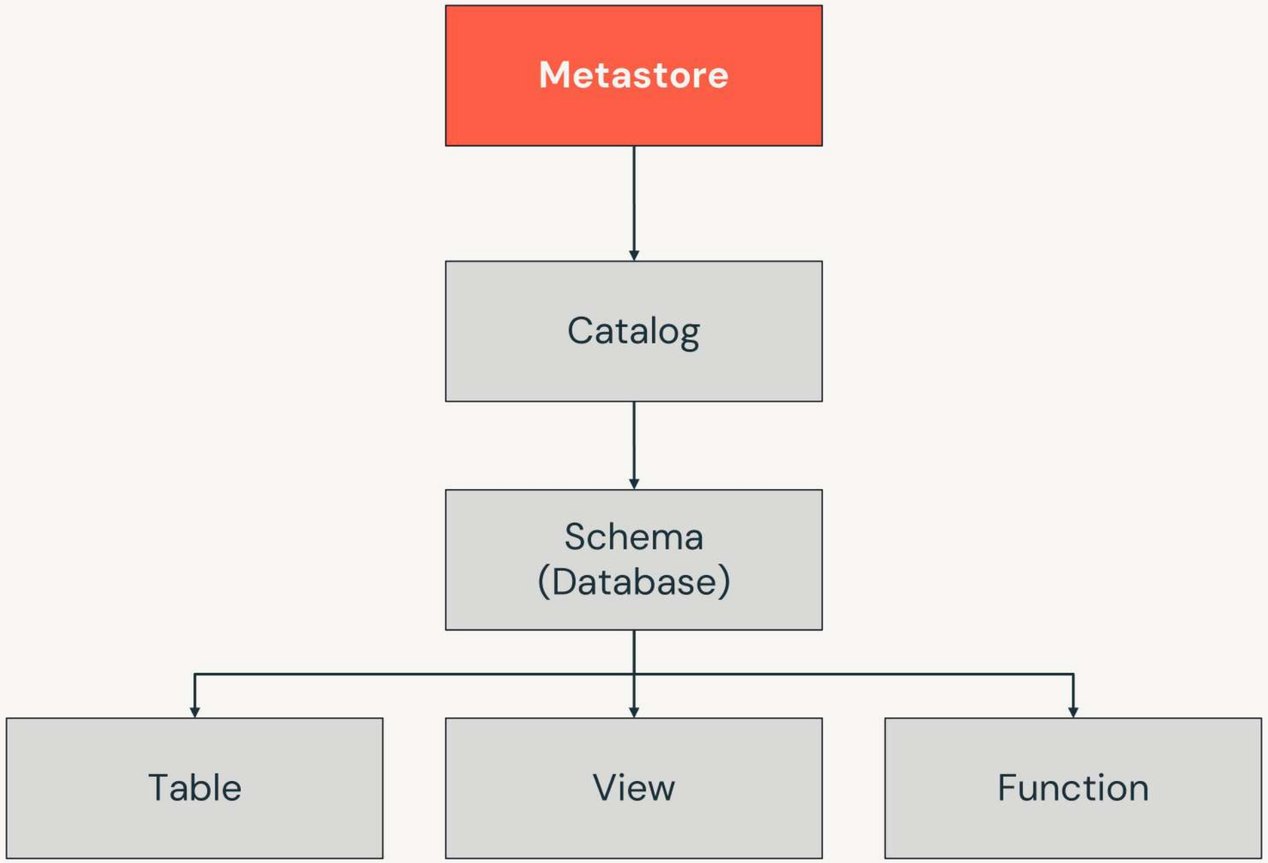
Data objects in the Lakehouse



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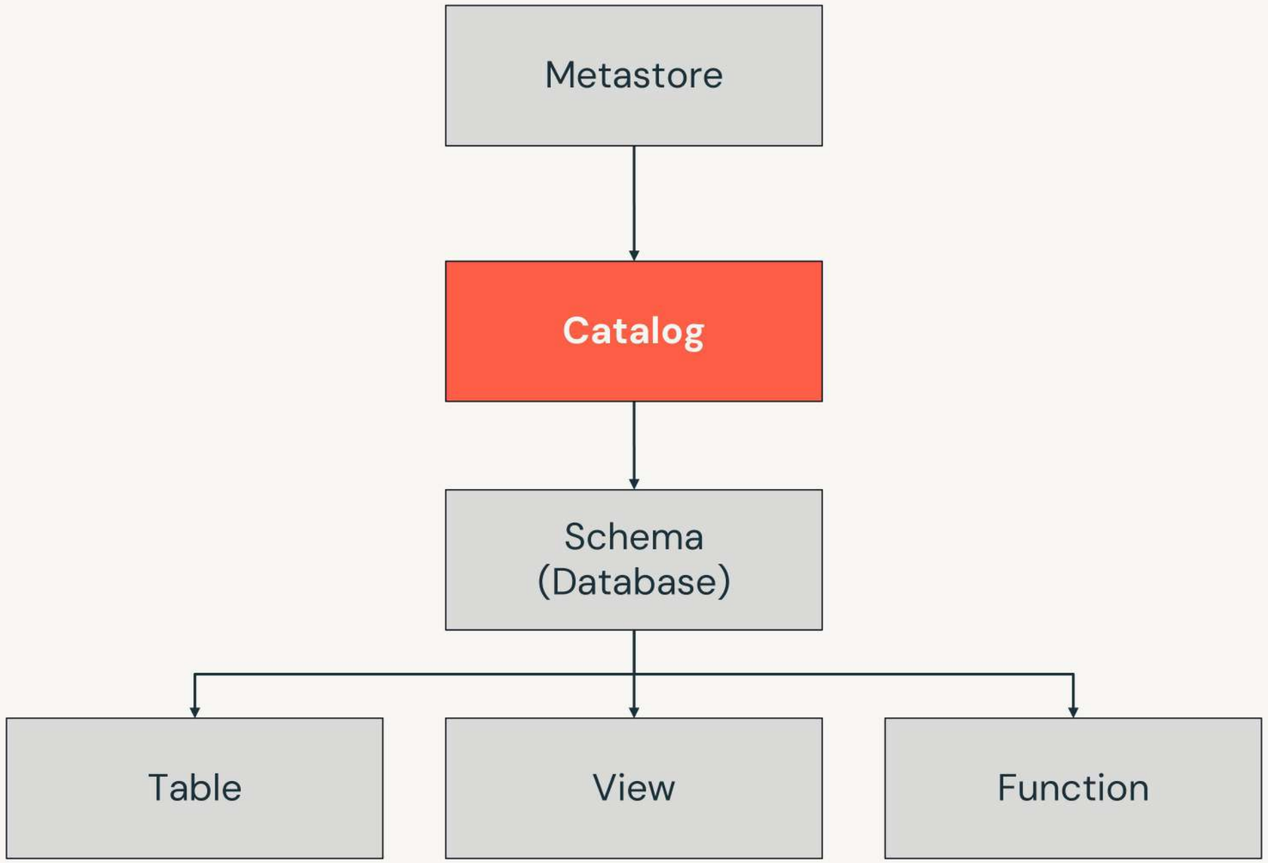
Data objects in the Lakehouse



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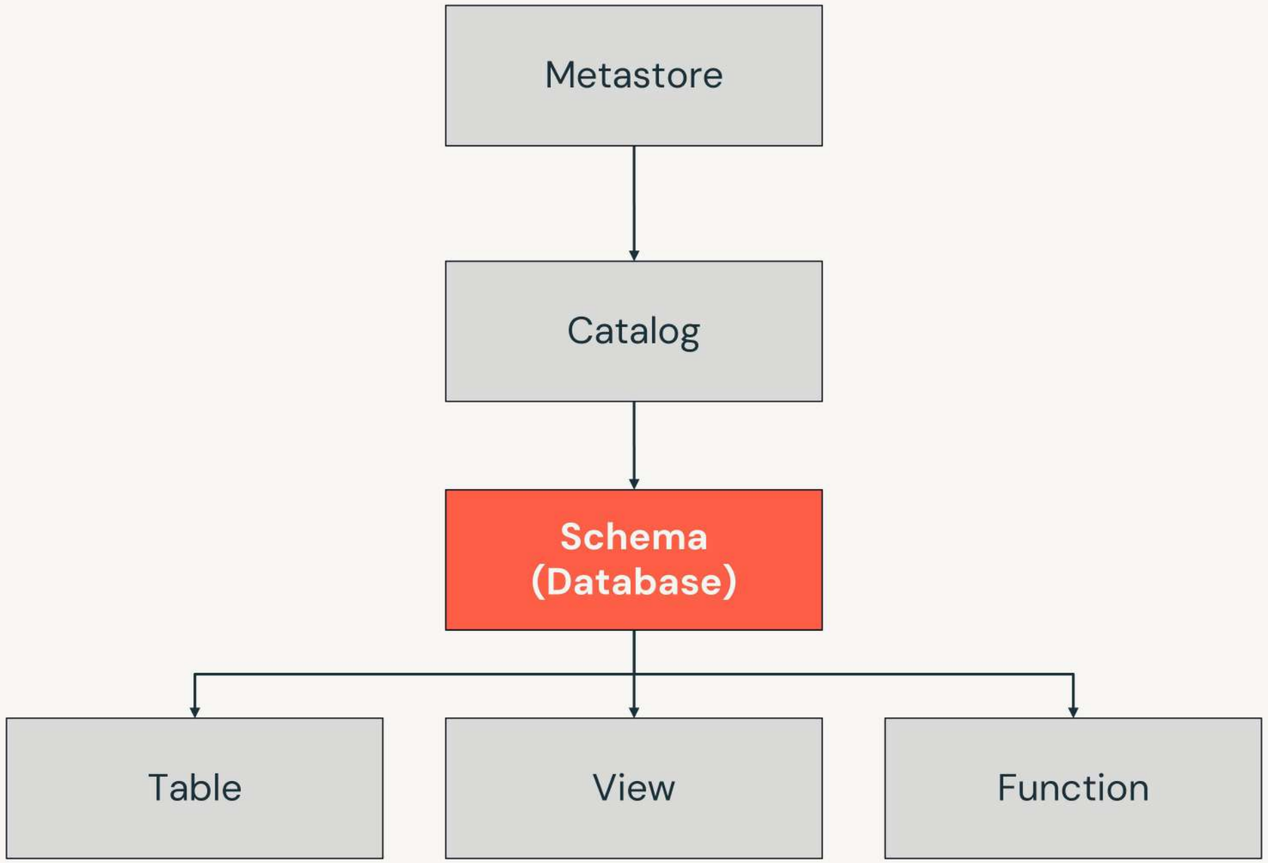

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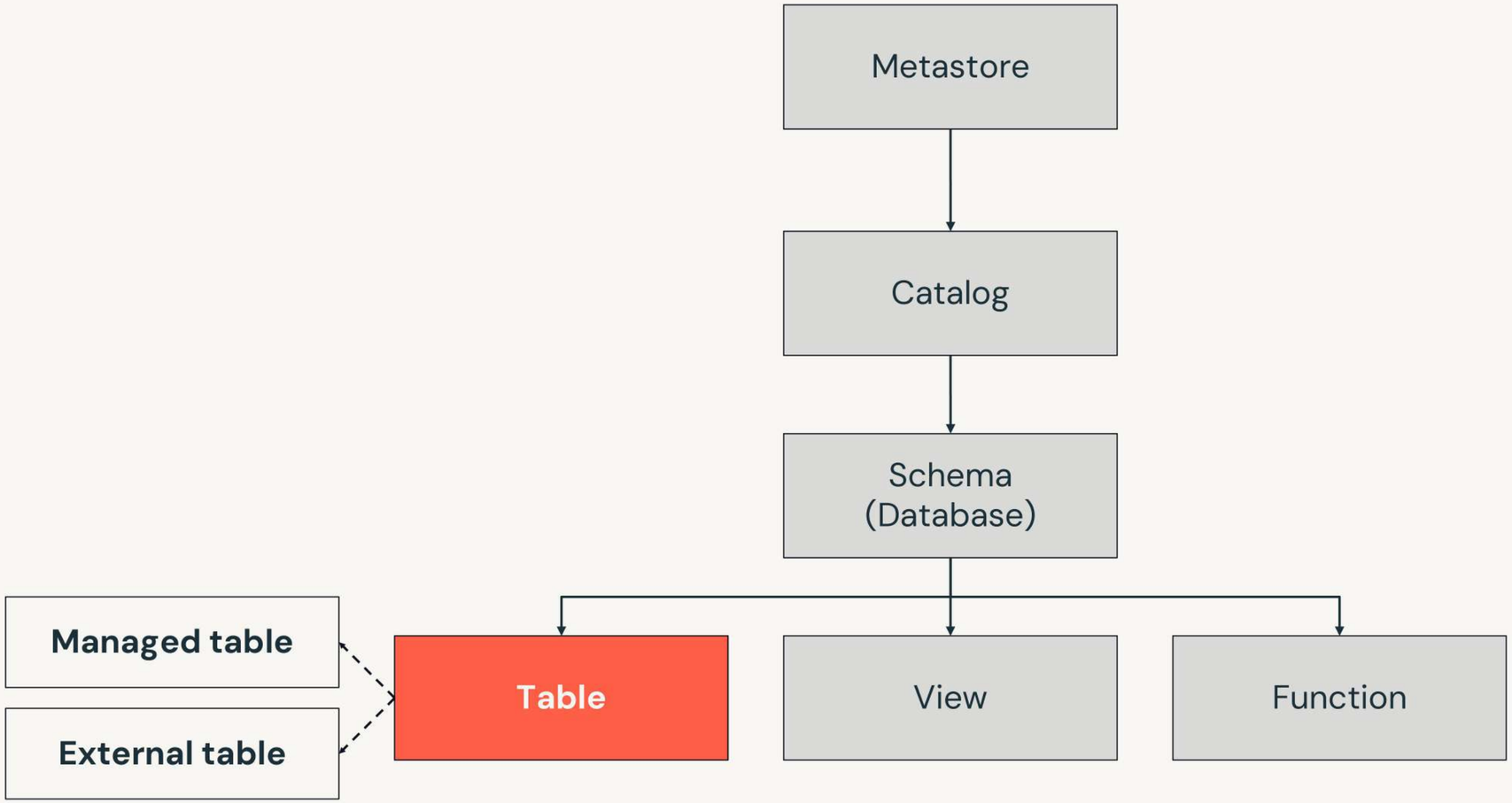

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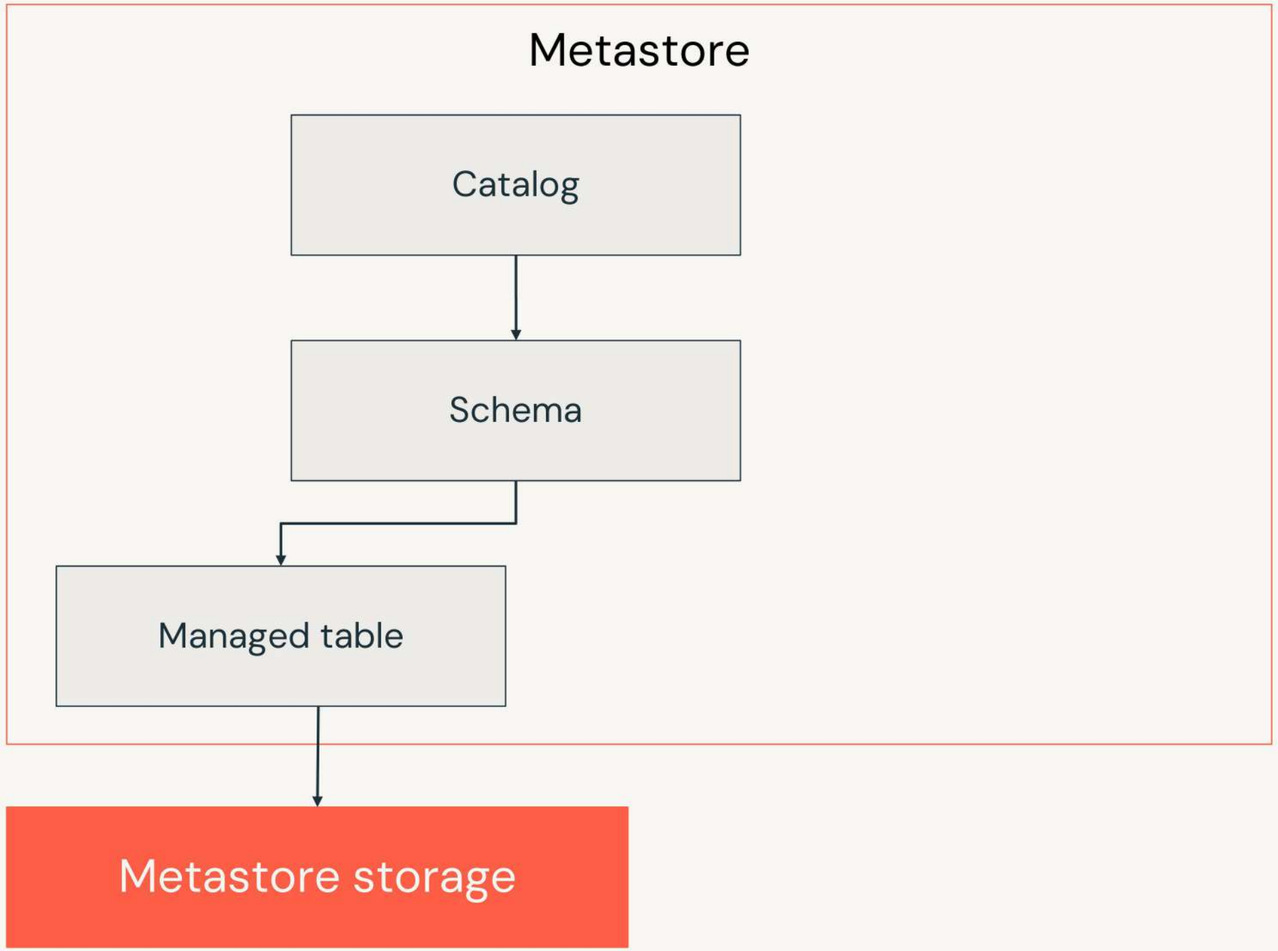
Managed Tables



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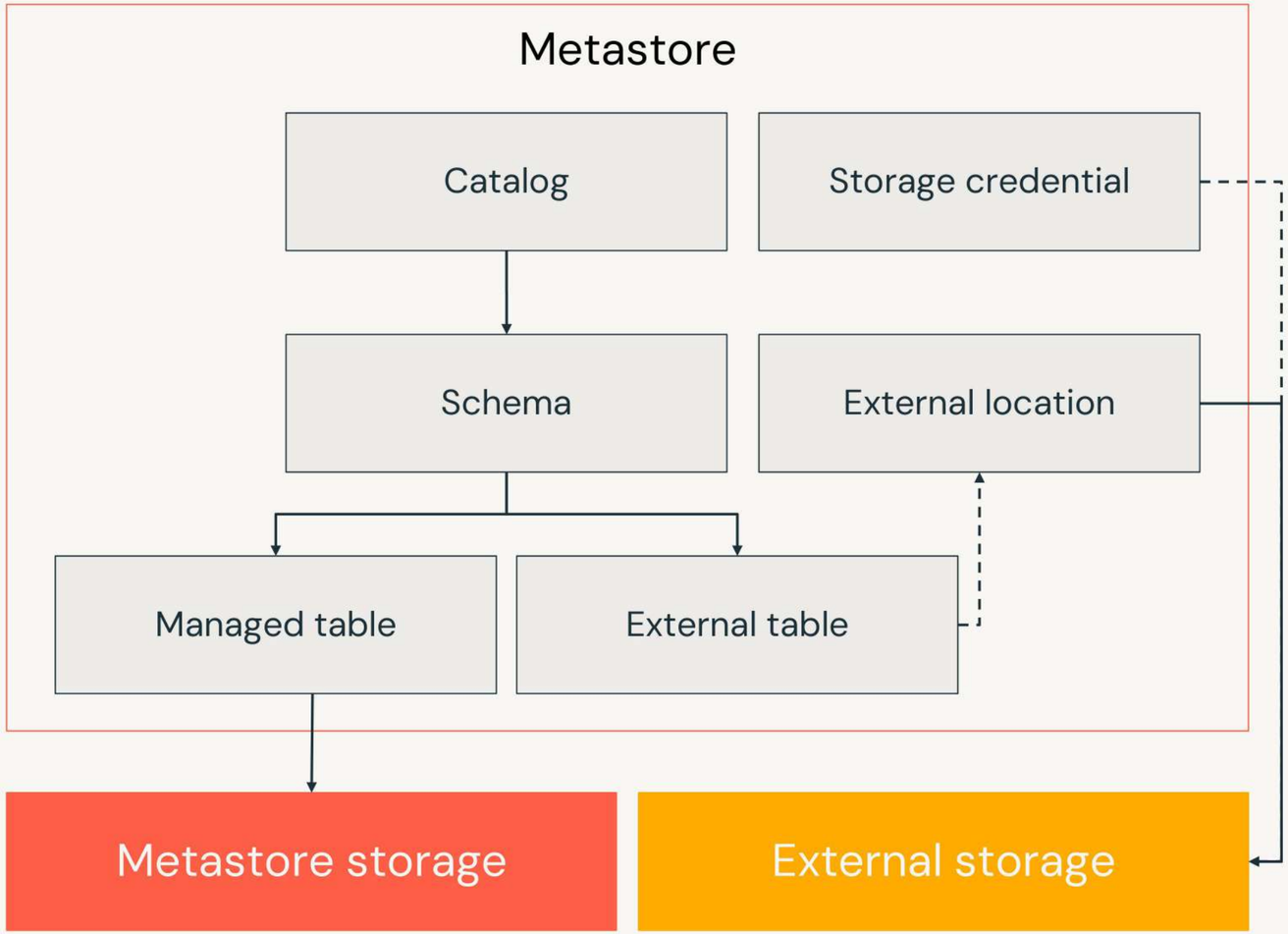
External Tables



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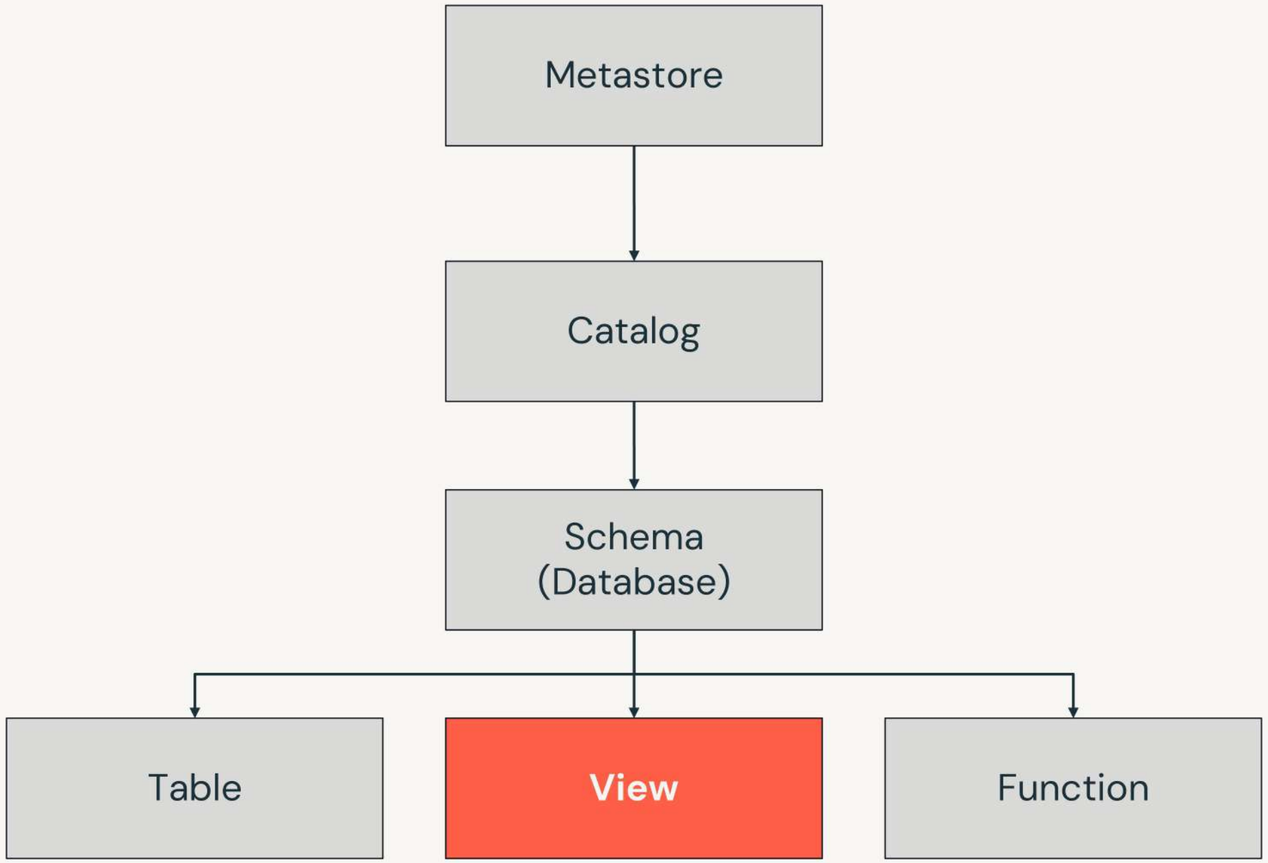
Data objects in the Lakehouse



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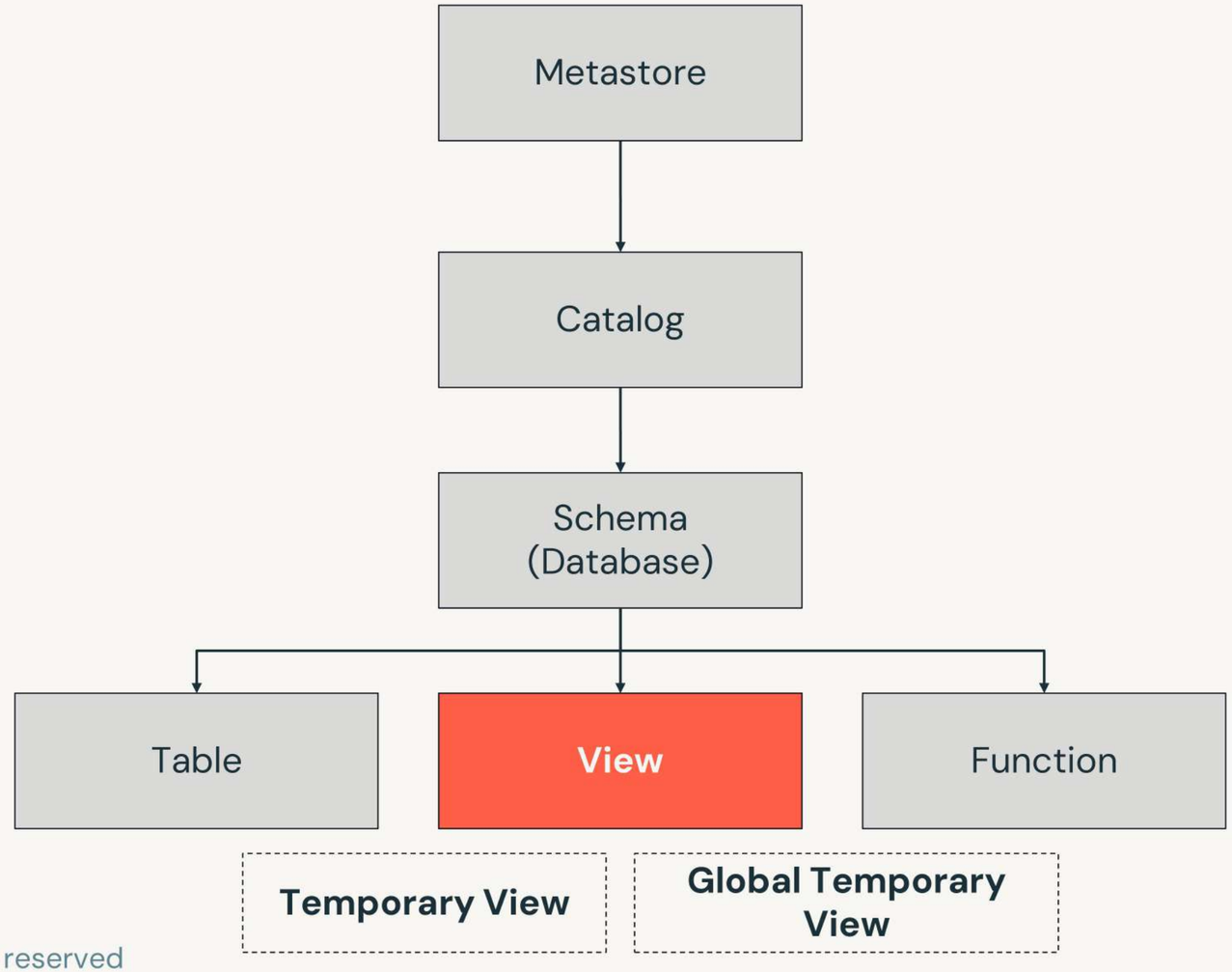

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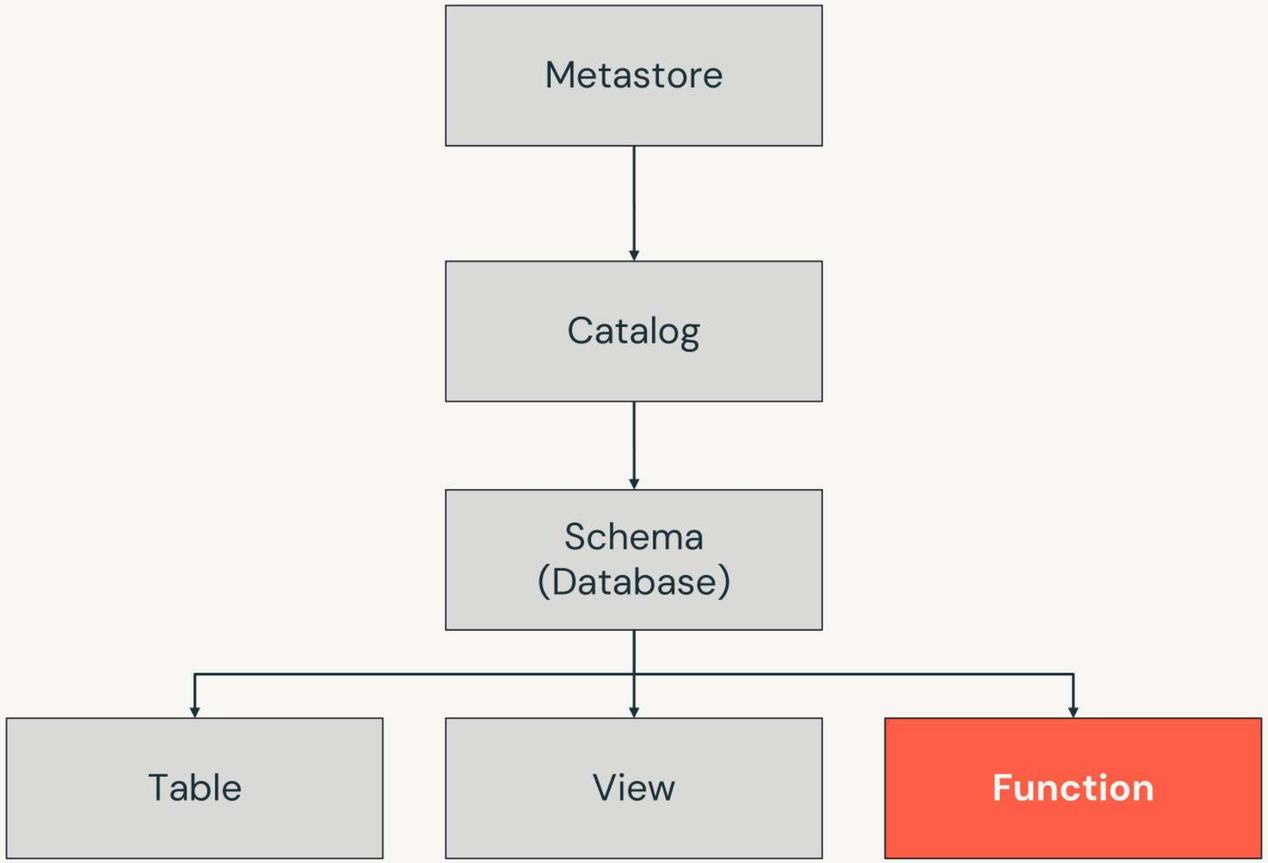
Data objects in the Lakehouse



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DE 2.1: Querying Files Directly

Use Spark SQL to directly query JSON data files

Leverage `text` and `binaryFile` methods to review raw file contents

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DE 2.2: Providing Options for External Sources

Use Spark SQL to configure options for extracting data from external sources

Create tables against external data sources for various file formats

Describe behavior when querying tables defined against external RDBMS sources

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DE 2.3L: Extract Data Lab

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DE 2.4: Cleaning Data

Summarize datasets and describe NULL behaviors

Retrieve and removing Duplicates

Validate datasets for expected counts, missing values, and duplicate records

Apply `date_format` and `regexp_extract` to clean and transform data

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DE 2.5: Complex Transformations

Use : and . syntax to traverse nested data in strings and structs

Use .* syntax to flatten and query struct types

Parse JSON string fields

Flatten/unpack arrays and structs

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DE 2.5L: Reshape Data Lab (Optional)

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DE 2.7A: SQL UDFs and Control Flow (Optional)

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DE 2.7B: Python UDFs (Optional)

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